

The Nature
Conservancy

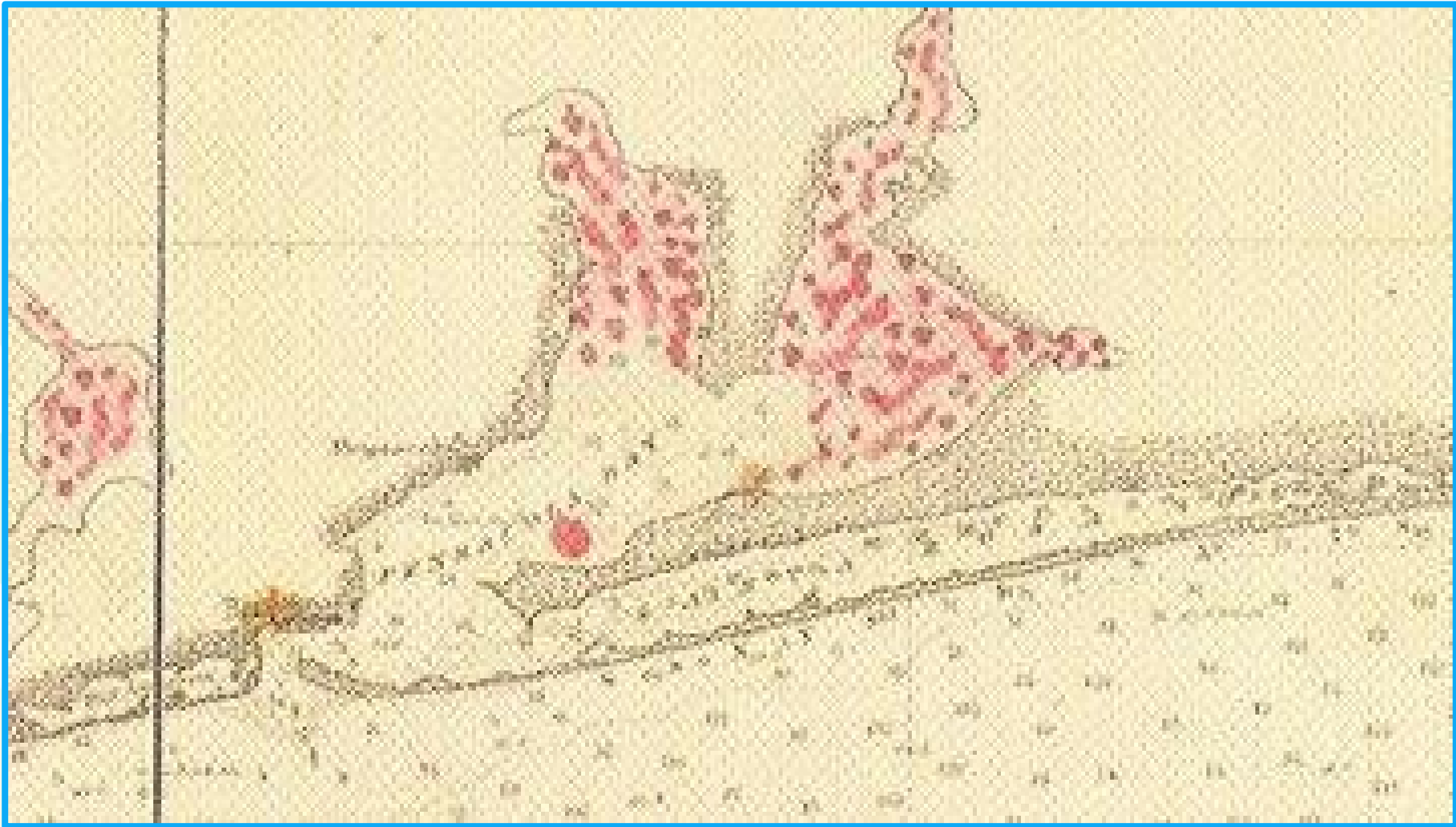


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ESTUARY PROGRAM

Oyster Habitat Suitability Model for Pensacola Bay

June 13, 2023

Pensacola Bay Historical Oyster Reefs

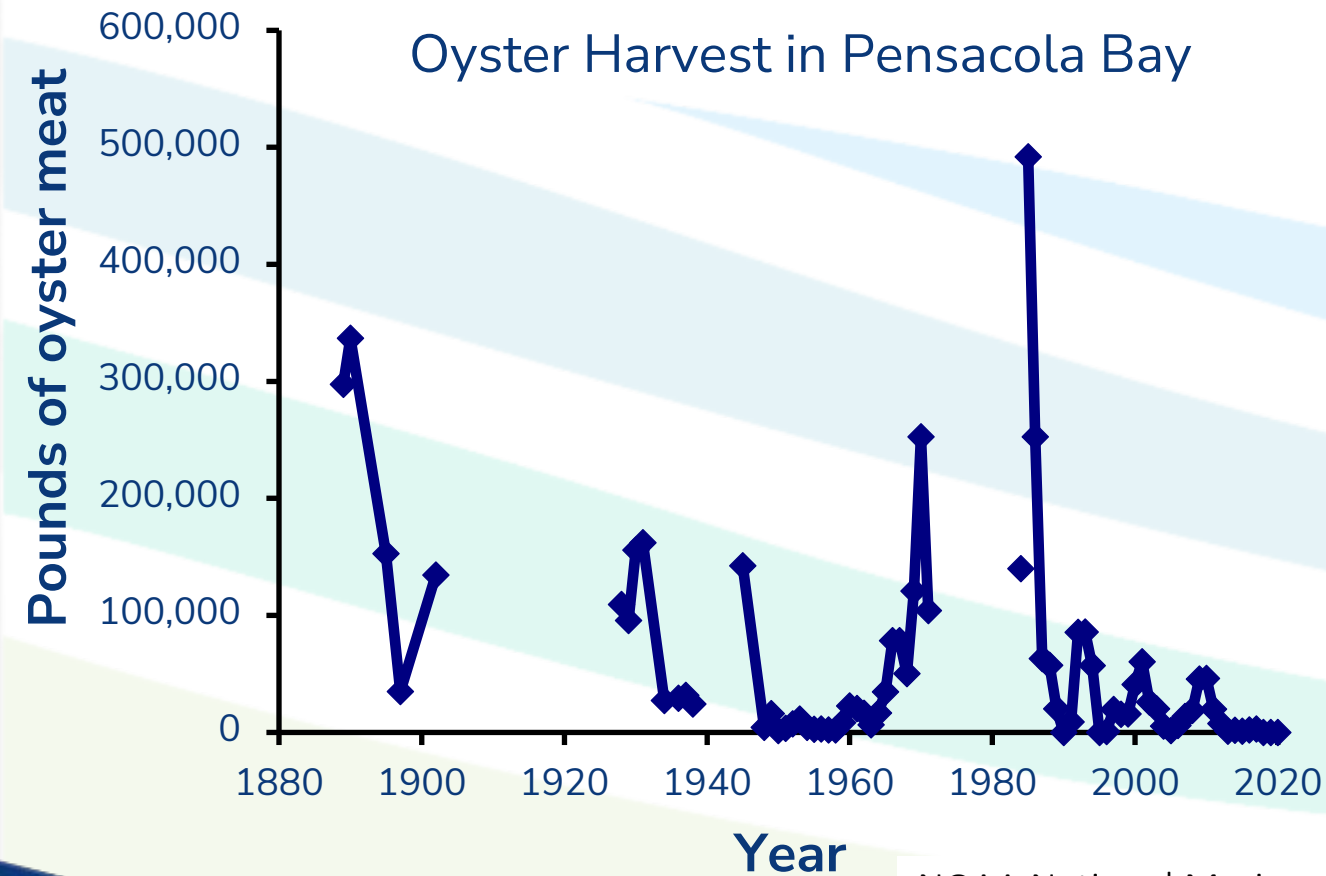


US Fish Commission 1883

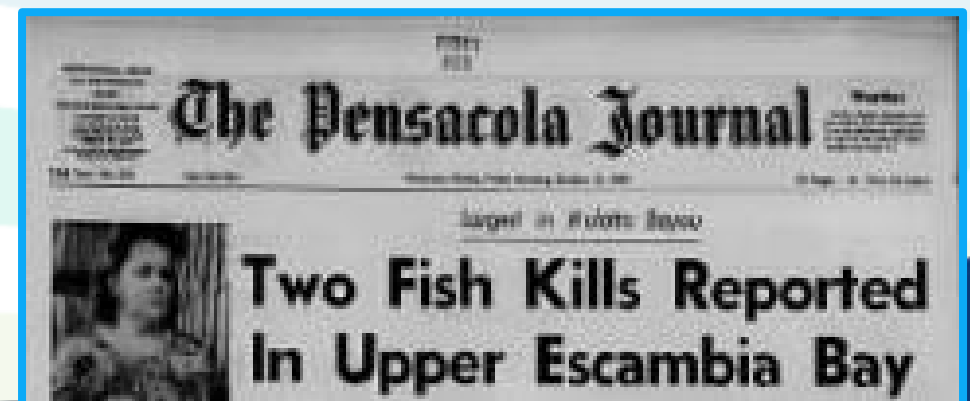


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Pensacola Bay Oysters



NOAA National Marine
Fisheries Service



October 1969

Roadmap to Recovery for Pensacola Bay

- Collaboratively developed with:
 - Oyster harvesters and farmers, resource managers, researchers, state/federal agencies, community organizations, etc.
- Bay-scale restoration plan
- Restore ecological function, the fishery, & habitat
- PPBEP's Oyster-Sub Committee to guide implementation

The Nature Conservancy 



OYSTER FISHERIES
AND HABITAT
MANAGEMENT PLAN
for the
Pensacola Bay System
May 2021



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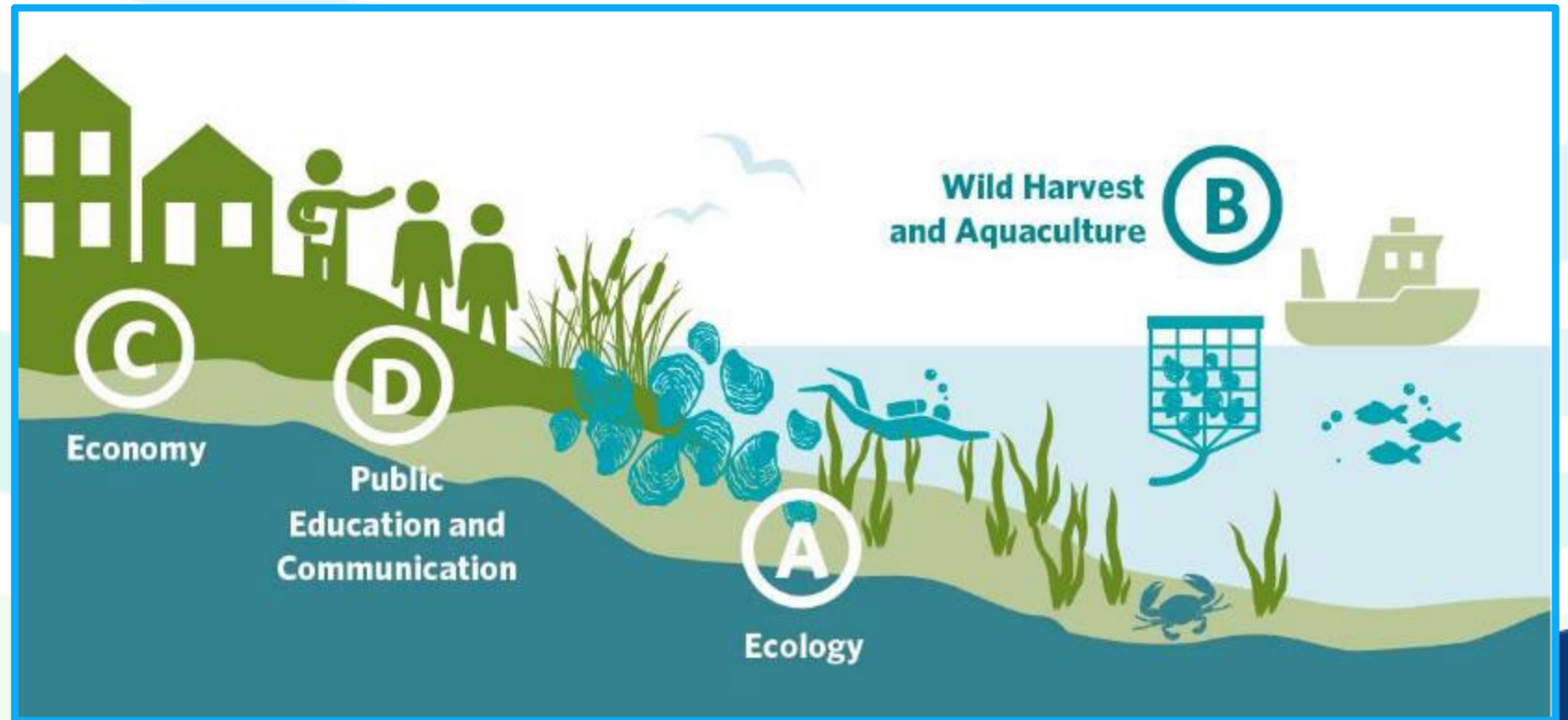
Birch et al. 2021 (TNC)

Implementing the Oyster Plan

CCMP Goal 5 Conserve and restore critical habitats



Objective 5.1: Improve and support health and resilience of oysters including wild, farmed, and restored oysters to promote a sustainable fishery in the Pensacola Bay System.



Steps Taken to Inform Reef Locations

- Oyster Reef Mapping
- Humble Oyster Documentary
- Water Filtration Model
- Habitat Suitability Model
- Vision Mapping



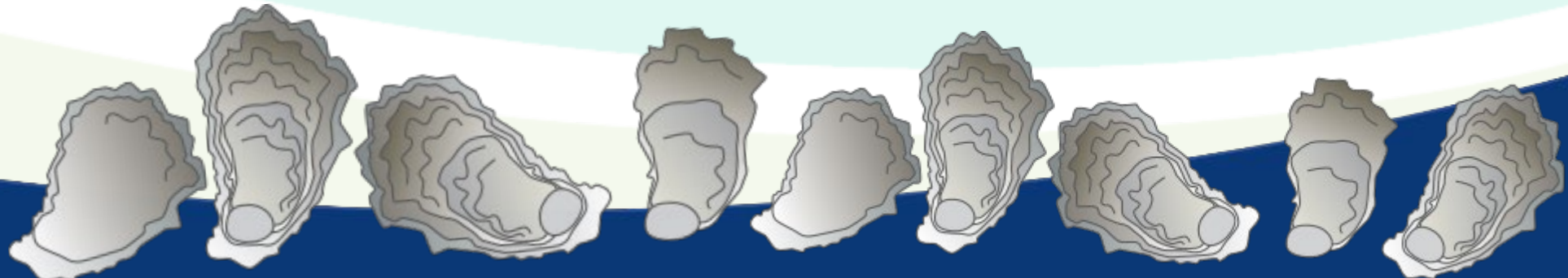
Oyster Habitat Suitability Model

Purpose:

Develop a habitat suitability model to better inform where restored oyster reefs should be placed based on areas with environmental conditions most suitable for oysters to live

Goal:

Co-develop criteria for prioritizing restoration for multiple objectives



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Methods

- Identify factors that affect oyster habitat suitability
- Compile and analyze data
- ‘Score’ locations within PBS and for each data layer from 0.00-1.00
- Combine scores from all data layers for a composite HSM index from 0.00-1.00
 - Closer to 0.00 = unsuitable
 - Closer to 1.00 = suitable



Metrics Considered

- Salinity
- Dissolved oxygen
- Substrate type
- Oyster recruitment
- Chlorophyll-a
- Temperature
- Water Flow
- Current oyster reef extent
- Historical oyster reef extent
- Submerged aquatic vegetation
- Disease, predation
- Sea Level Rise
- Shoreline type
- Water Quality



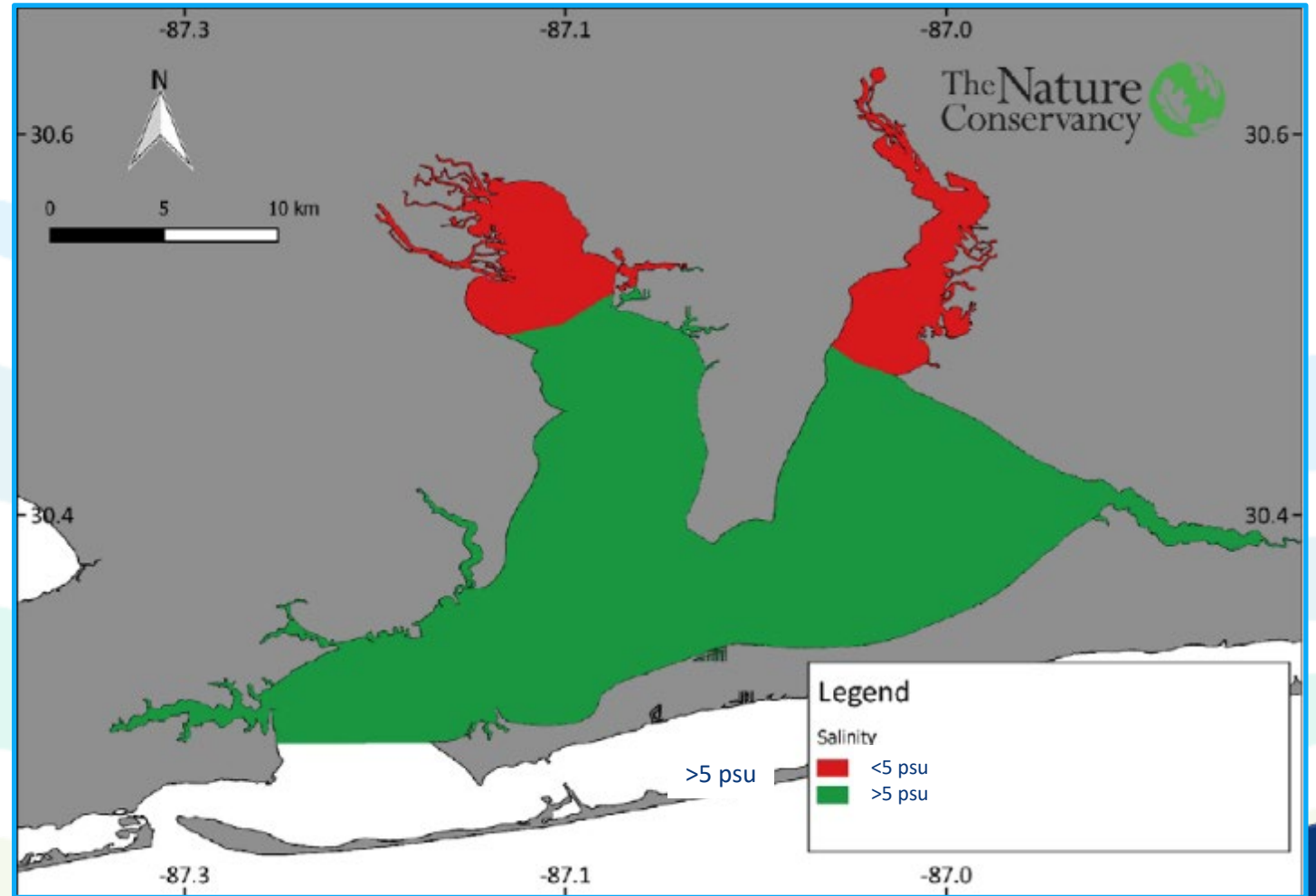
Metrics Included

Metric	Unit	Reference	Model Scoring
Current oyster reefs	Presence	Fish and Wildlife Research Institute	Reefs Present = 1 Reefs Absent = 0
Historical oyster reefs	Presence	U.S. Fish Commission	Reefs Present = 1 Reefs Absent = 0
Dissolved oxygen (bottom, summer)	mg/L	USEPA 2005	≥ 2 mg/L = 1 < 2 mg/L = 0
Seagrass	Presence	Fish and Wildlife Research Institute	Seagrass Present = 0 Seagrass Absent = 1
Substrate type	Type	USEPA 2005	Mud = 0 Muddy Sand = 0.25 Sand = 0.5
Salinity	psu	USEPA 2005	≥ 5 = 1 < 5 = 0.5
Recruitment	Spat/shell	Arnold et al. 2017	Variable from 0-1



Salinity

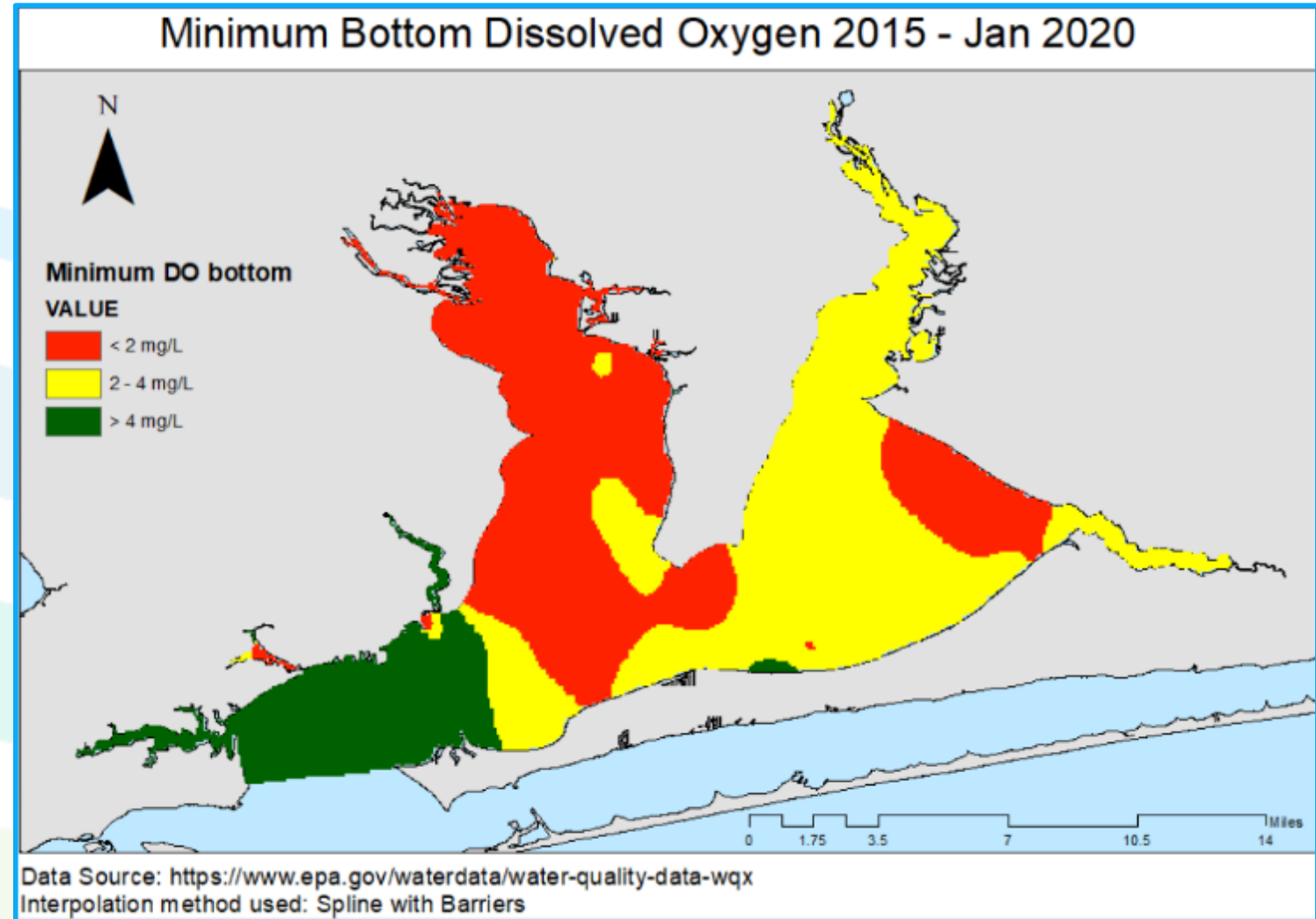
- **Salinity range: 1-5 PSU**
- Data from USEPA 2005 using EPA's Environmental Monitoring and Assessment Program
- Quarterly Sampling
- 1995-2000



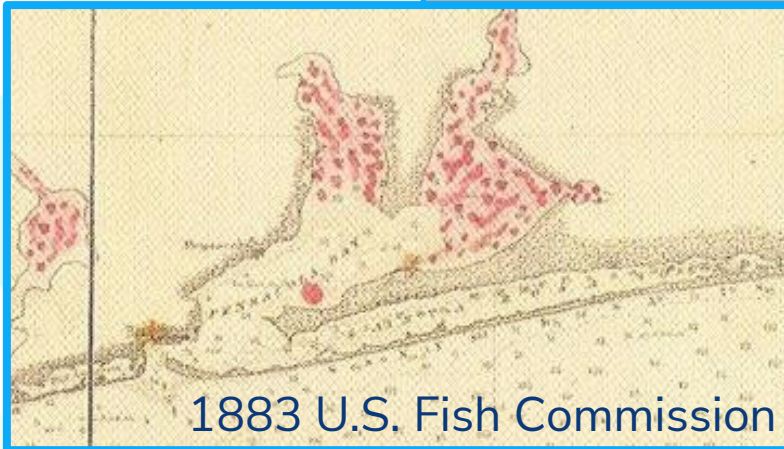
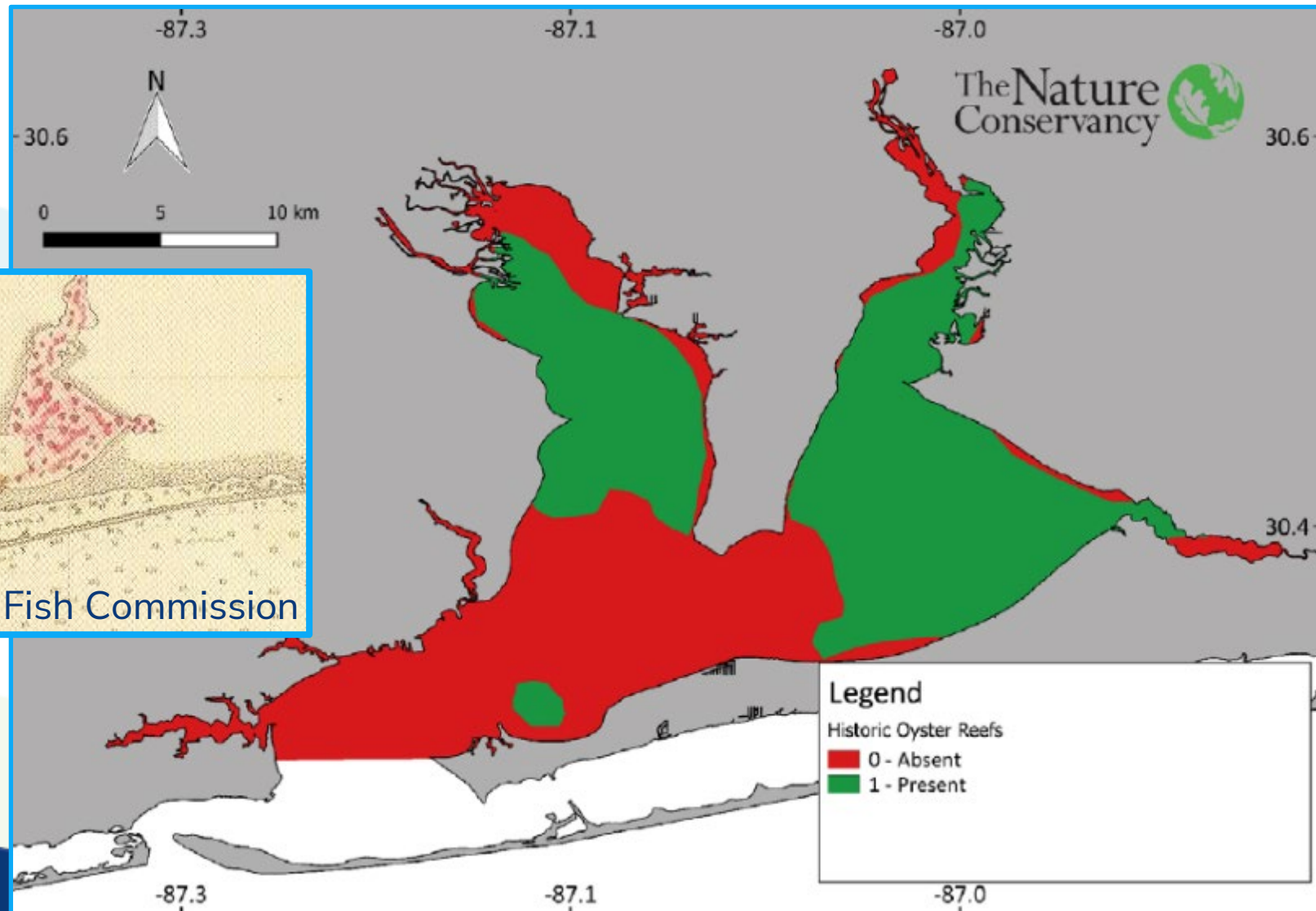
Dissolved Oxygen

DO range:
<2 mg/L
2-4 mg/L
>4 mg/L

- Data from USEPA 2005 using EPA's Environmental Monitoring and Assessment Program
- Quarterly Sampling
- 1995-2000

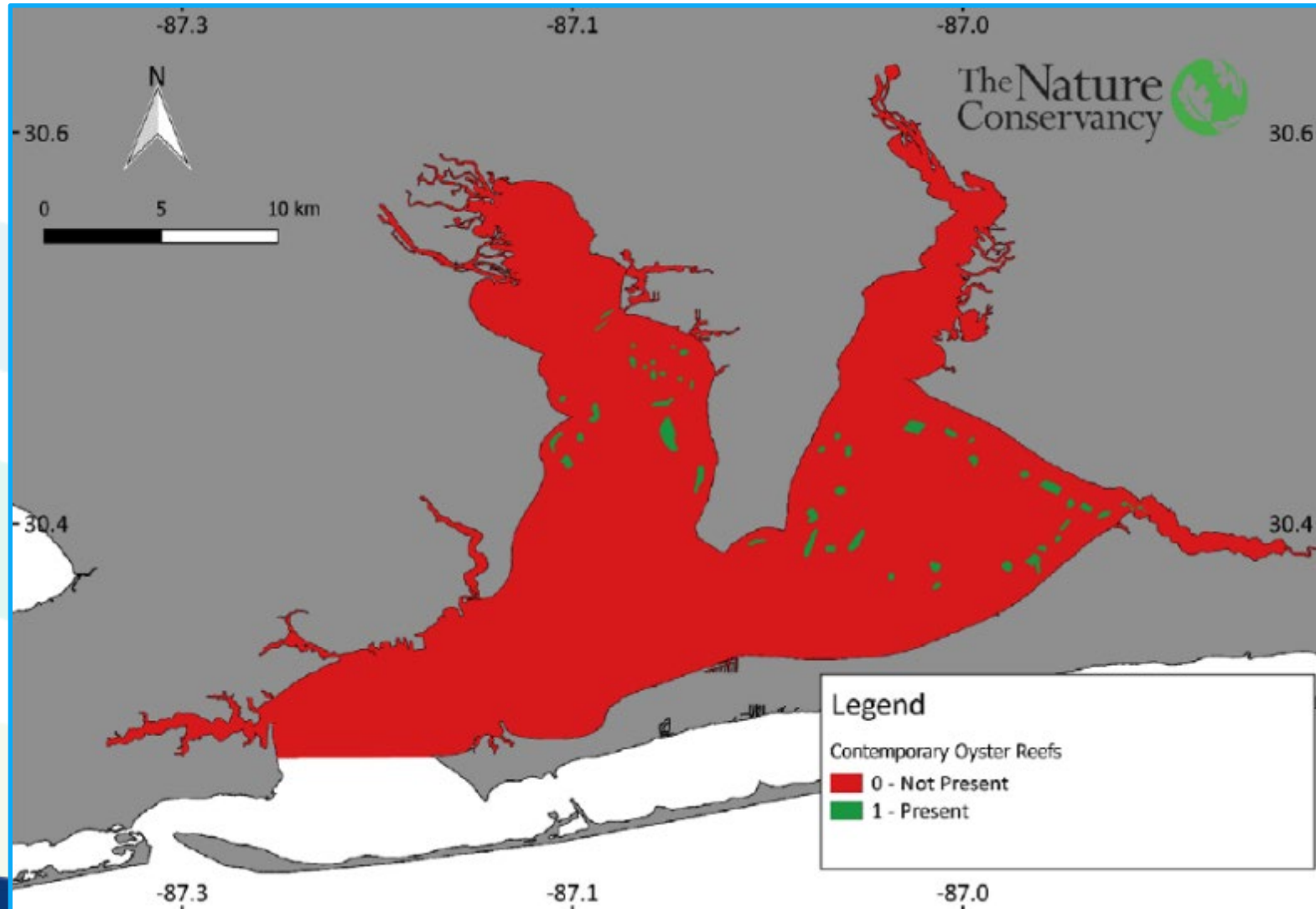


Historical Oyster Reefs



Current Oyster Reefs

2010 data from
Fish and Wildlife
Research Institute

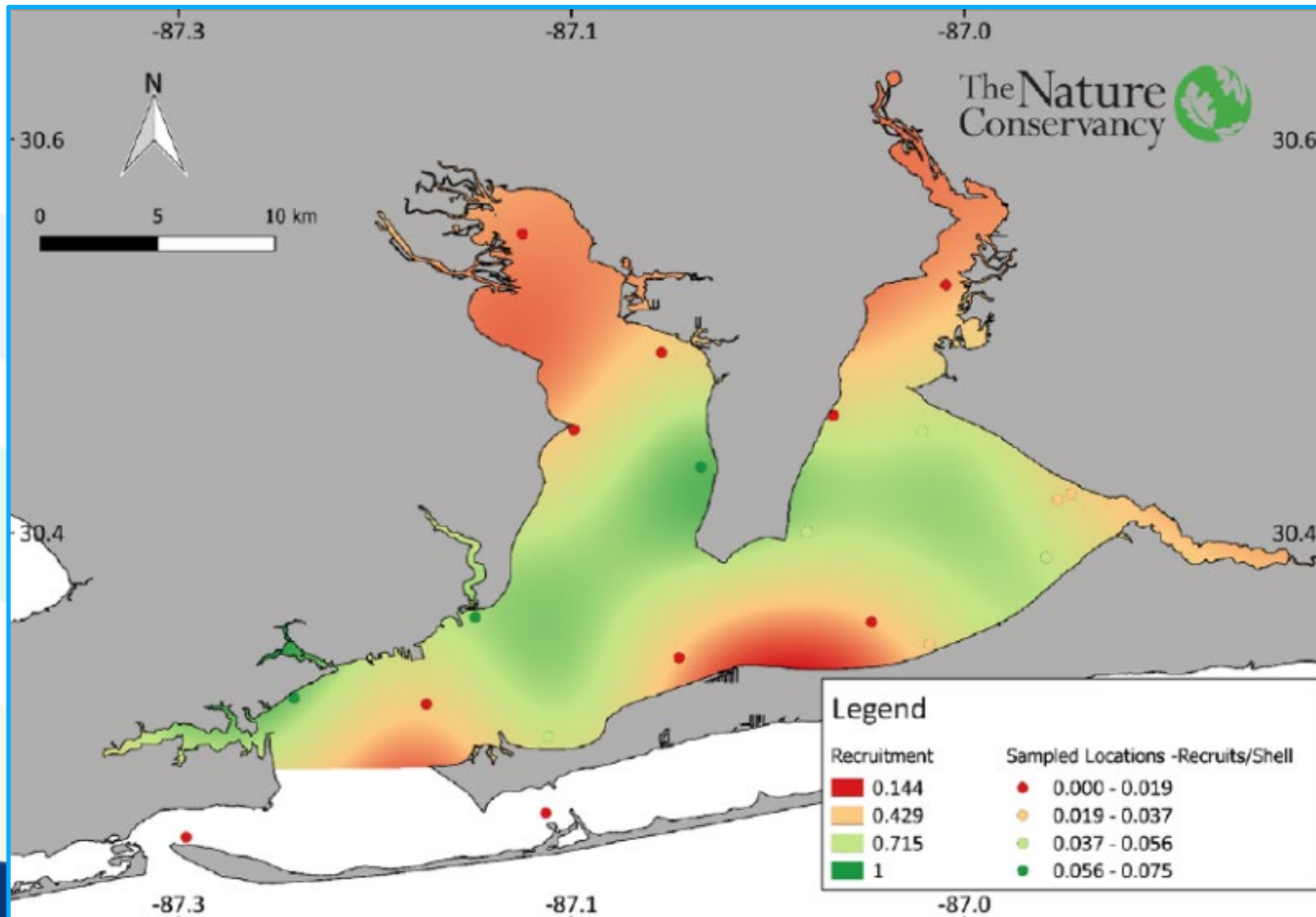


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Credit: Laura Geselbracht (TNC)

Recruitment

Arnold et al. 2017

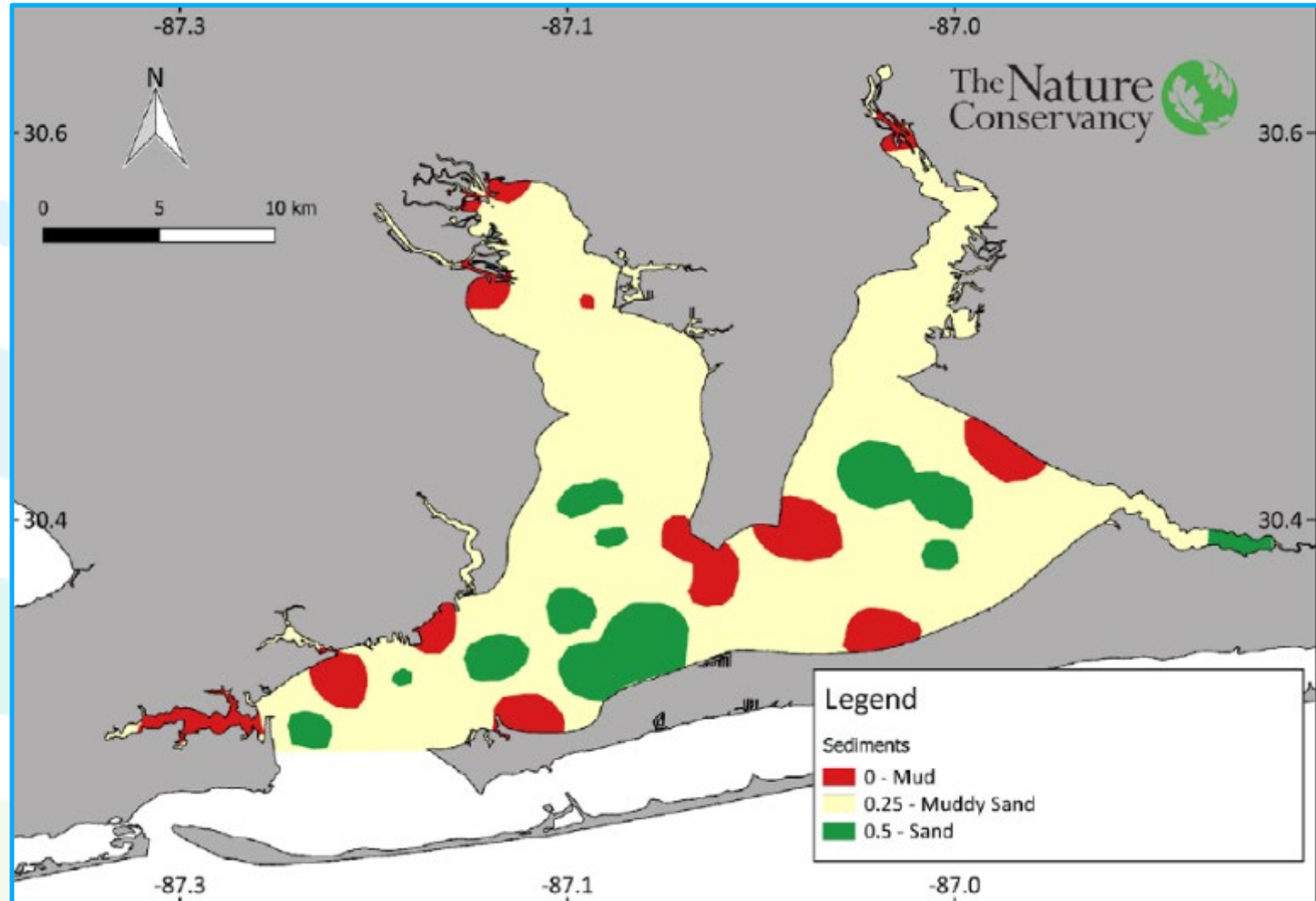


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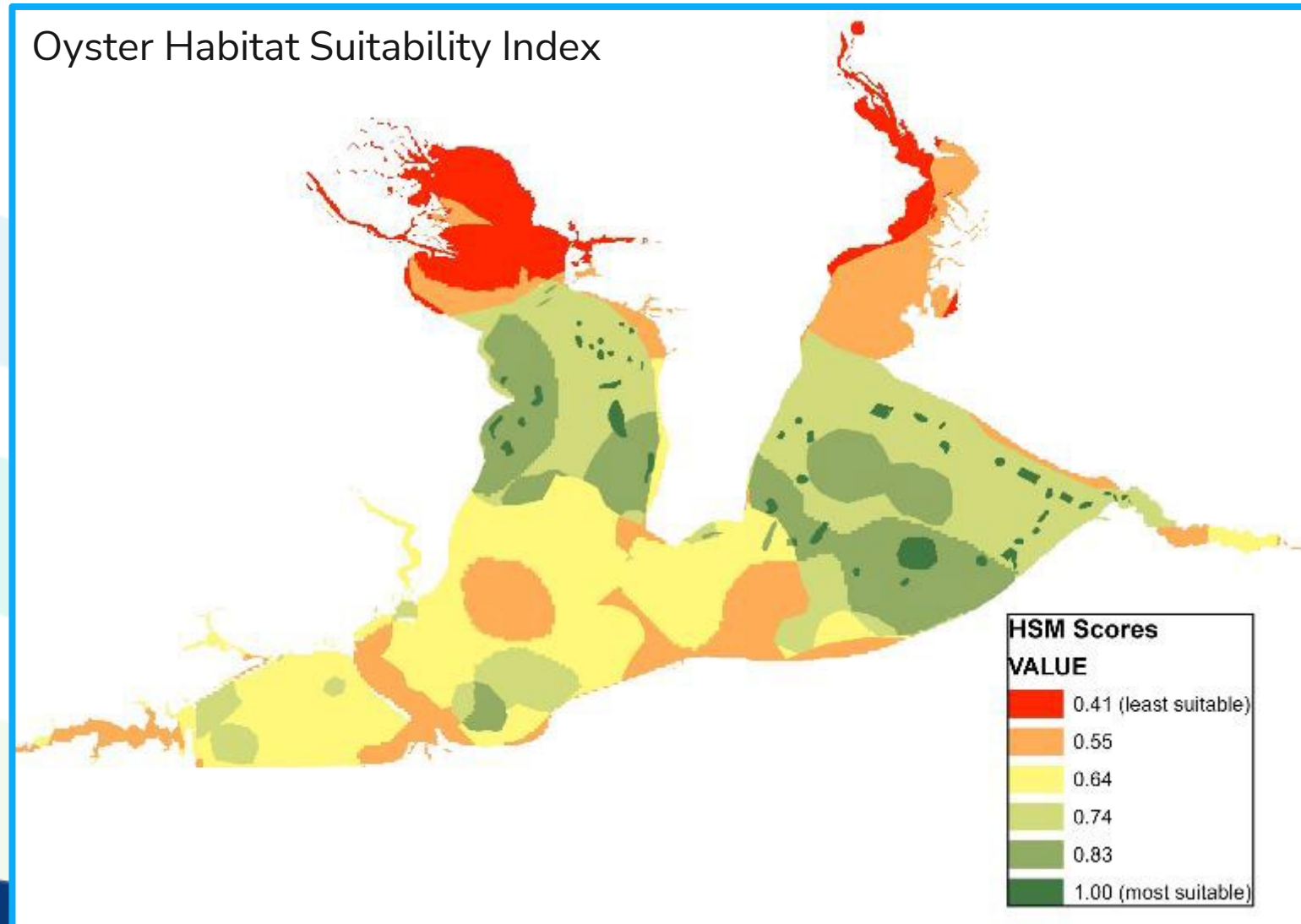
Credit: Laura Geselbracht (TNC)

Substrate Type

- Data from USEPA 2005 using EPA's Environmental Monitoring and Assessment Program
- Quarterly Sampling
- 1995-2000



Where is it most suitable for oysters?



2023 Model Update

Water Quality Portal Data (2000-2022)

- Salinity
- Dissolved oxygen
- Chlorophyll-*a*
- Turbidity

➤ Current oyster reefs –
2021 mapping

➤ ~~Submerged aquatic
vegetation~~

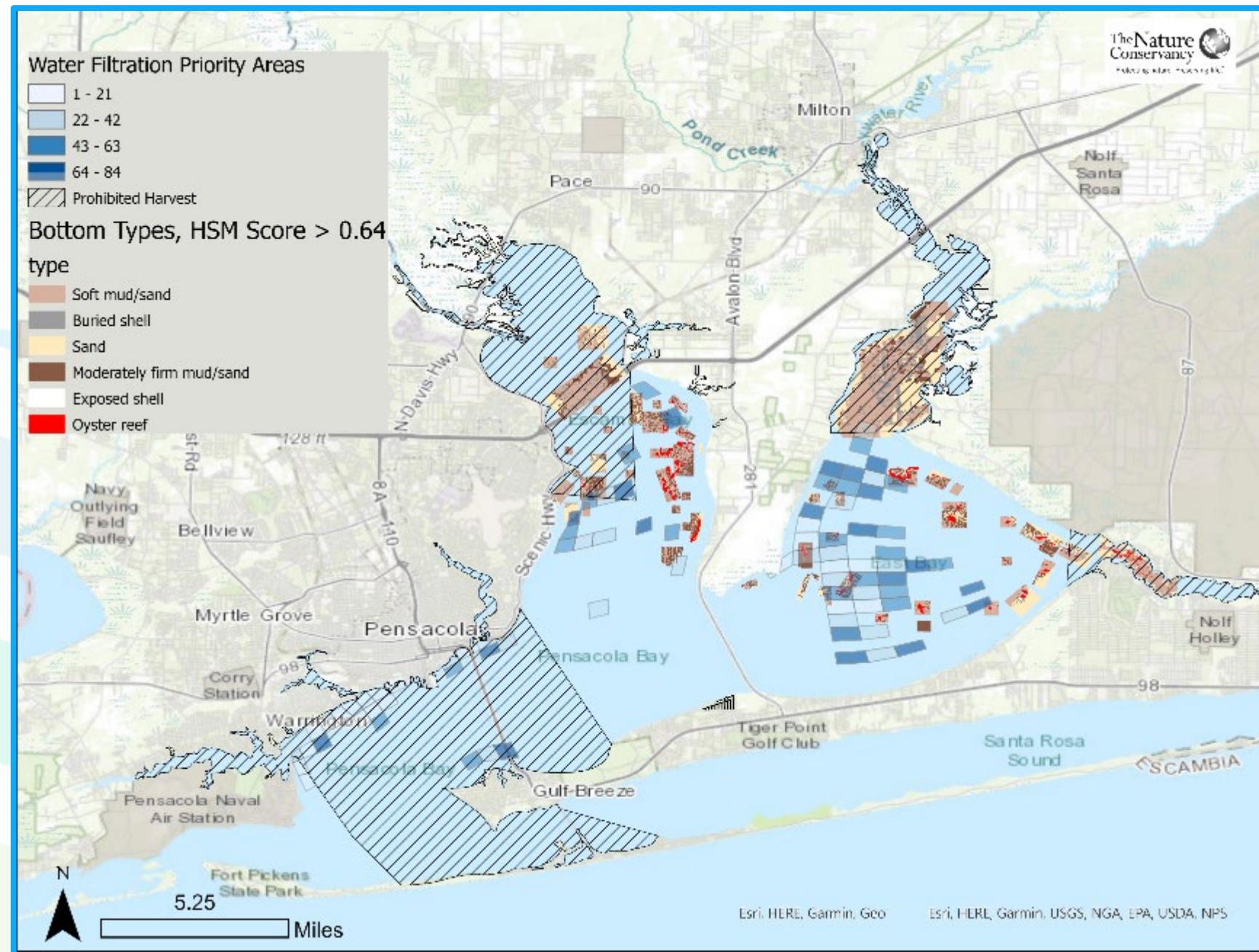
No change

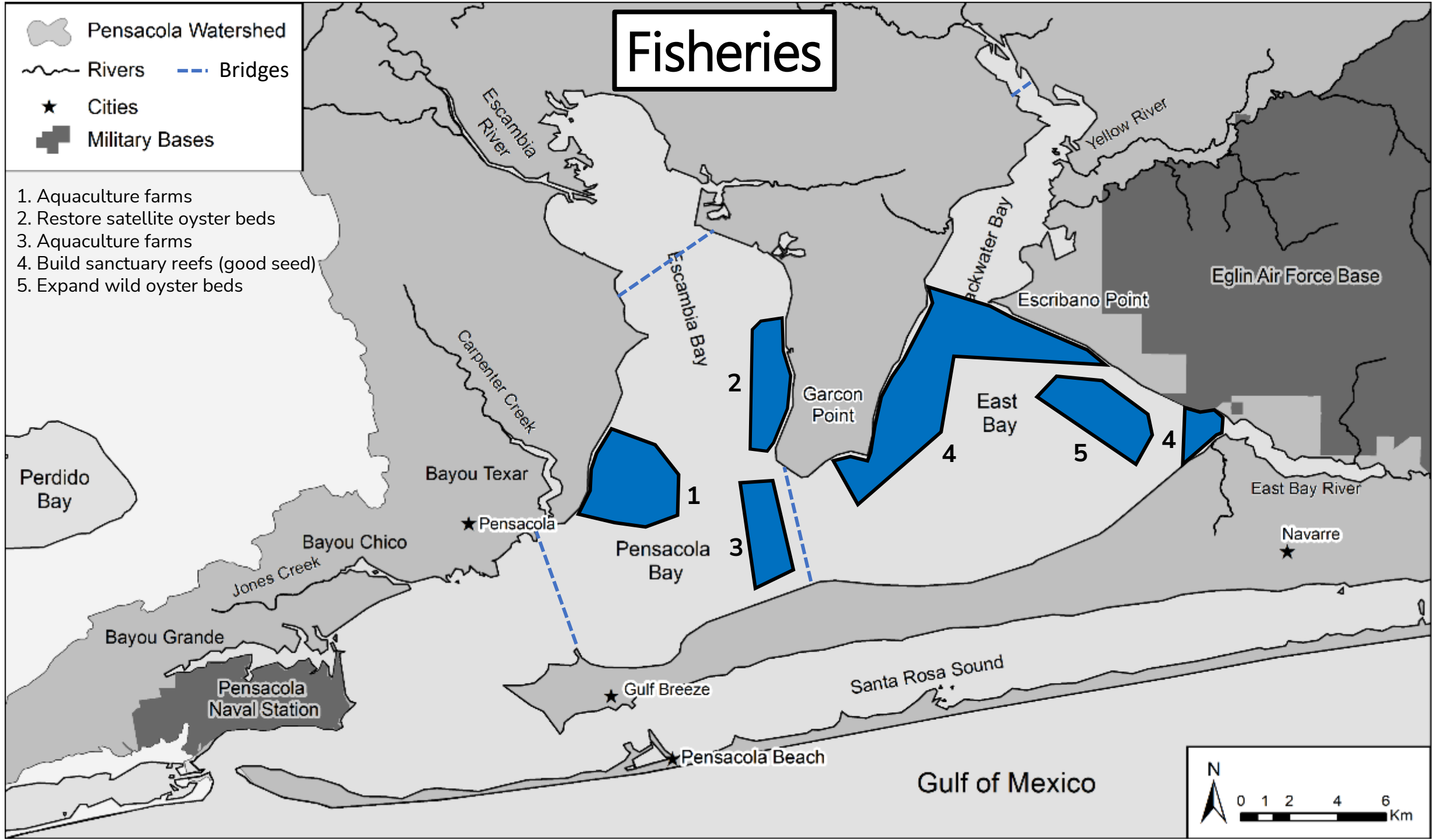
- Historical oyster reefs
- Substrate type
- Oyster recruitment

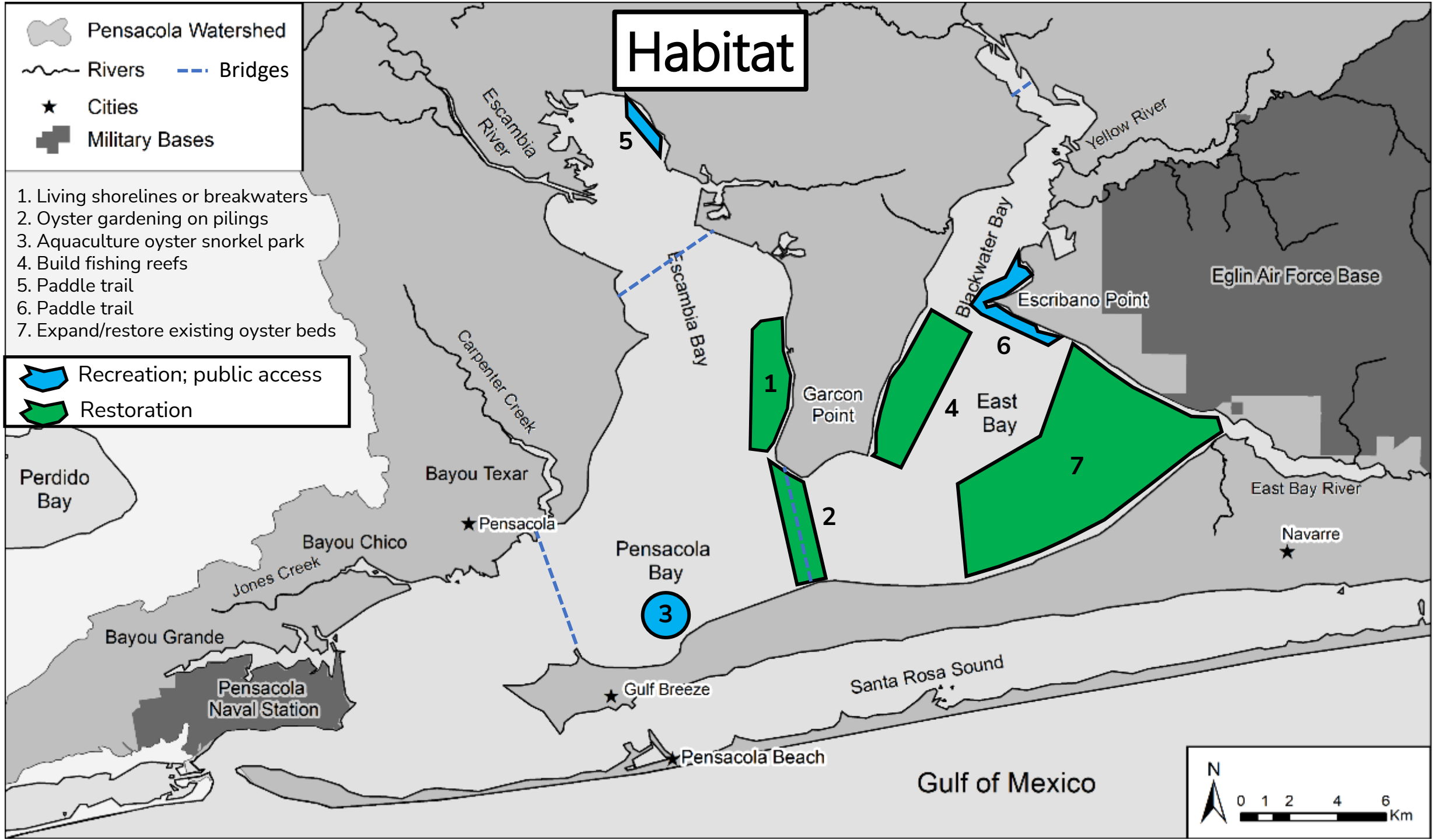


Bringing it all together!

- Environmental parameters
- Harvest classification
- Existing reefs
- Bottom types
- Water filtration priorities
- Larval supply



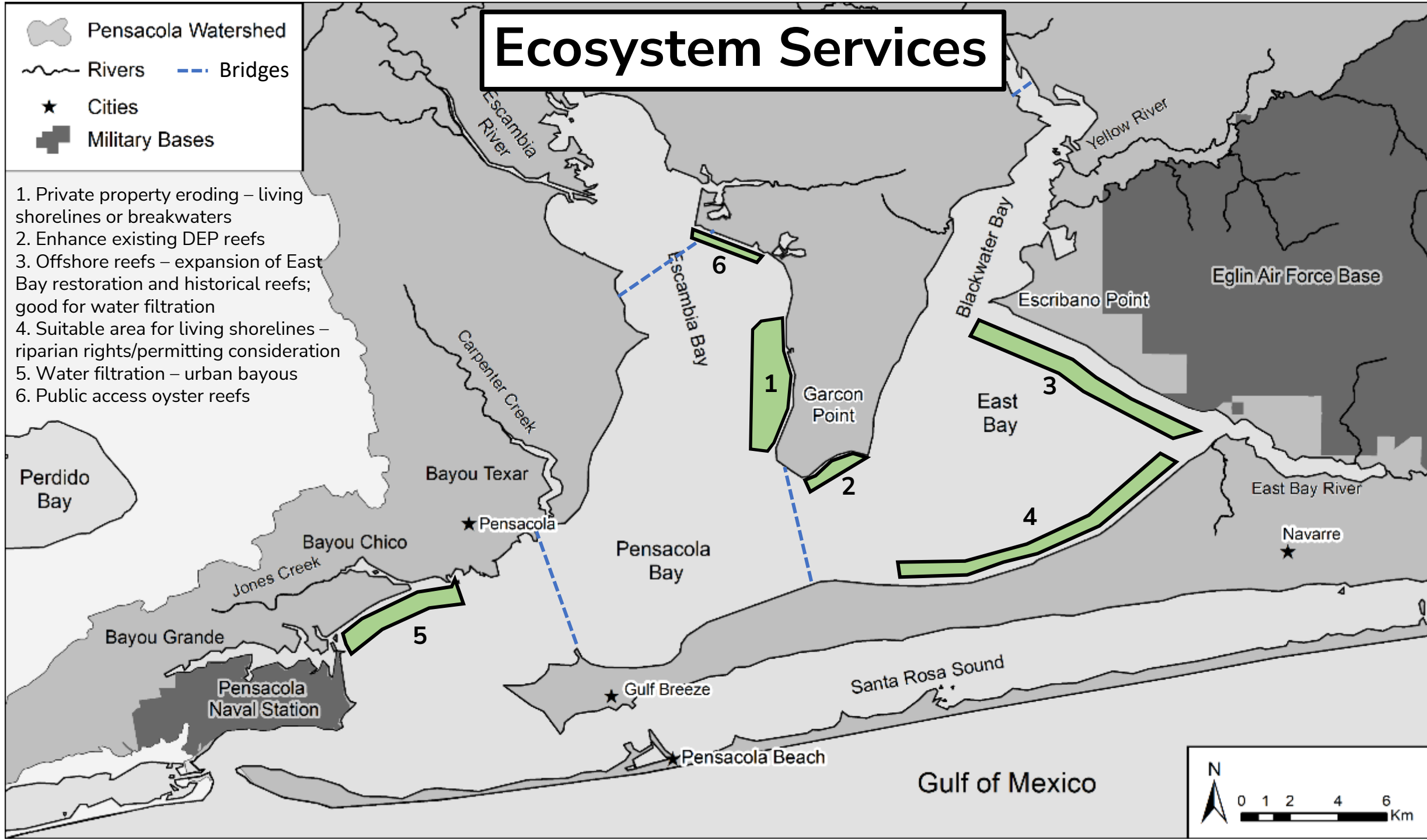




Ecosystem Services

-  Pensacola Watershed
-  Rivers
-  Bridges
-  Cities
-  Military Bases

1. Private property eroding – living shorelines or breakwaters
2. Enhance existing DEP reefs
3. Offshore reefs – expansion of East Bay restoration and historical reefs; good for water filtration
4. Suitable area for living shorelines – riparian rights/permitting consideration
5. Water filtration – urban bayous
6. Public access oyster reefs





Contact PPBEP



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Pensacola Bay Oyster Restoration Initiative

Funding Source: NOAA IJA (4 years; 2023-2026)

Lead: PPBEP

Request: \$11 M

- Partners: TNC, Santa Rosa County, FWC, DEP
- 5 Milestones:
 - Oyster restoration design and permitting
 - Oyster restoration Phase I implementation
 - Sediment load assessment
 - Sandy Hollow Gully restoration design and permitting
 - Living Shoreline Cost Share Program



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